DHCP Lab 1

Wednesday, 11 March, 2026 09:14 PM

Dhcp

<u>Title</u> : DHCP Configuration using Router in Cisco Packet Tracer

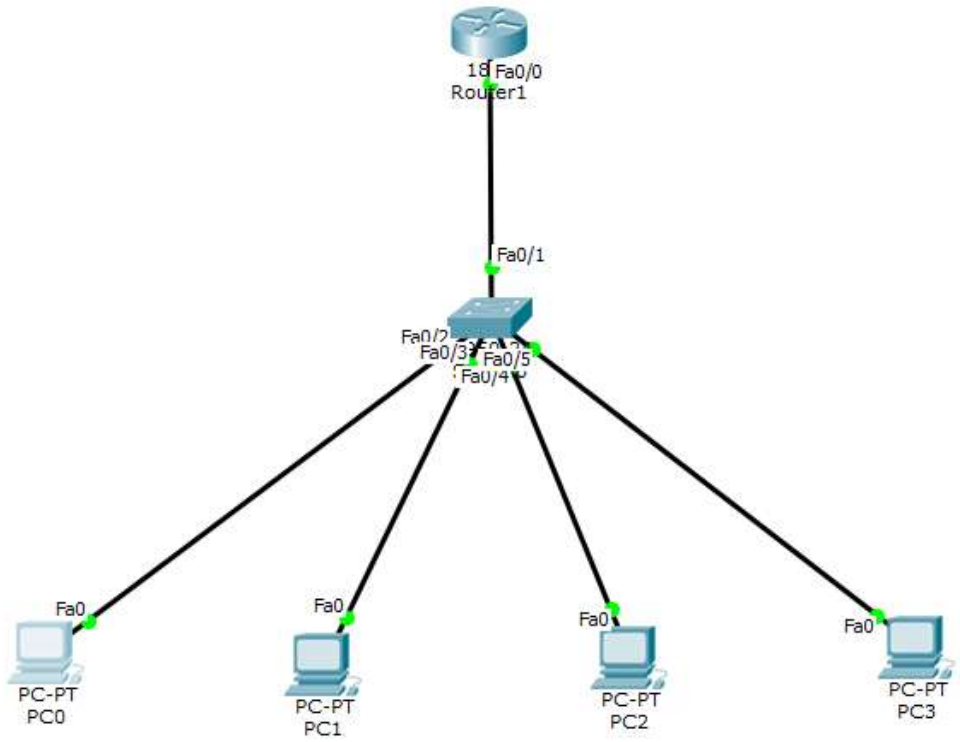
<u>Aim</u> : To configure a router as a DHCP server to automatically assign IP addresses to computers in a network.

Required Devices (Simulated Devices)

1. Router 1
2. Switch 1
3. PC 4

Topology Description

1. One router connected to a switch.
2. Four PCs connected to the switch.
3. The router acts as the DHCP server.



Network Addressing Scheme :

Device	Interface	IP Address	Subnet Mask
Router	Fa0/0	192.168.10.1	255.255.255.0
PC0	DHCP	Automatic	255.255.255.0
PC1	DHCP	Automatic	255.255.255.0
PC2	DHCP	Automatic	255.255.255.0
PC3	DHCP	Automatic	255.255.255.0

Network Address : 192.168.10.0/24

Procedure :

Step 1: Create Network Topology

1. Open Cisco Packet Tracer.
2. Add the following devices:
   - 1 Router
   - 1 Switch
   - 4 PCs
3. Connect the devices using Copper Straight Through cables.

   Connections:

From	**To**
Router Fa0/0	Switch F0/1
PC0	Switch F0/2
PC1	Switch F0/3
PC2	Switch F0/4
PC3	Switch F0/5

Step 2: Configure Router Interface

Open router CLI and enter:

→ enable
→ configure terminal
→ interface g0/0
→ ip address 192.168.10.1 255.255.255.0
→ no shutdown
→ exit

This assigns the router as the default gateway.

Step 3: Exclude IP Addresses

Exclude the first five IP addresses so DHCP will not assign them.

→ ip dhcp excluded-address 192.168.10.1 192.168.10.5

Excluded IPs:
192.168.10.1
192.168.10.2
192.168.10.3
192.168.10.4

192.168.10.5

Step 4: Configure DHCP Pool

→ ip dhcp pool TECHSOLUTIONS
→ network 192.168.10.0 255.255.255.0
→ default-router 192.168.10.1
→ dns-server 8.8.8.8
→ Exit

Step 5: Configure PCs to Obtain IP Automatically

For each PC:

- Click PC
- Desktop
- IP Configuration
- Select DHCP

Each PC will automatically receive an IP address.

Verification

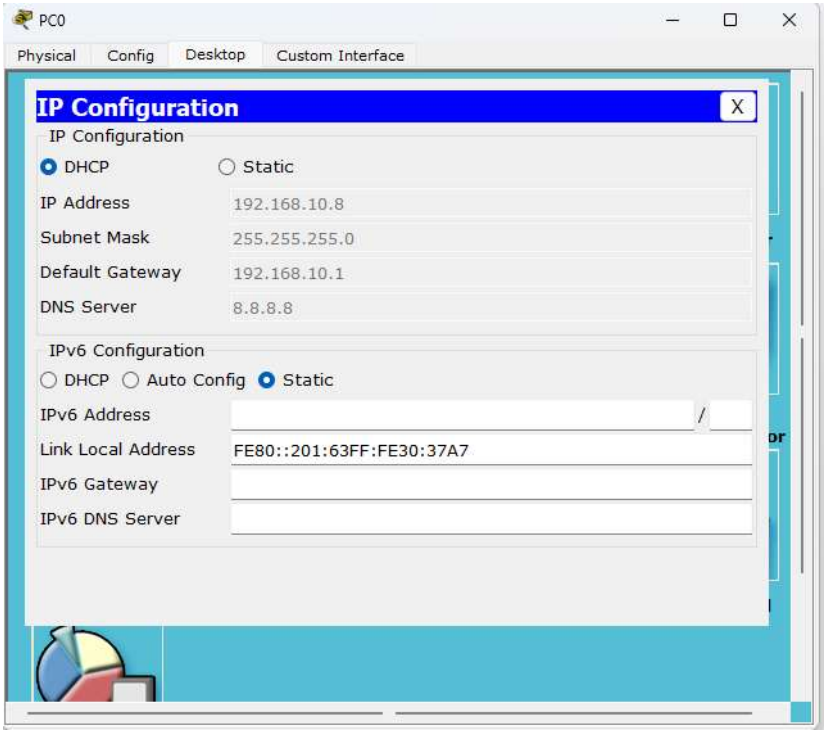
Check IP Address

Open PC Command Prompt:

→ ipconfig

Example Output:

PC	IP Address	Gateway
PC0	192.168.10.6	192.168.10.1
PC1	192.168.10.7	192.168.10.1
PC2	192.168.10.8	192.168.10.1
PC3	192.168.10.9	192.168.10.1


PC0
Physical
Config
Desktop
Custom Interface
IP Configuration
X
IP Configuration
DHCP
Static
IP Address
192.168.10.8
Subnet Mask
255.255.255.0
Default Gateway
192.168.10.1
DNS Server
8.8.8.8
IPv6 Configuration
DHCP
Auto Config
Static
IPv6 Address
Link Local Address
FE80::201:63FF:FE30:37A7
IPv6 Gateway
IPv6 DNS Server